

CHONG TANG WAI, NIEL

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PROFESSIONAL SUMMARY

Backend software engineer with production experience in Java, Spring Boot, and Kotlin at OCBC, complemented by active machine learning development in Python using scikit-learn and PyTorch. Experienced in building end-to-end ML pipelines covering statistical feature selection, multi-model comparison, and CI/CD orchestration, alongside a full-stack anomaly detection system with a FastAPI inference layer and React Native front-end. Additional depth in DevSecOps and infrastructure automation. Targeting backend or AI-integrated engineering roles at fintechs and financial institutions.

SKILLS

Languages: Java · Python · Kotlin · SQL · Shell Scripting · PowerShell · HTML/CSS

Frameworks & APIs: Spring Boot · Spring Security · Spring Data JPA · REST APIs · FastAPI · React Native

ML / AI: PyTorch · NumPy · Pandas · Neural Networks · Time-Series Modelling · Anomaly Detection · Scikit-learn · Gradient Boosting · Random Forest · Logistic Regression · Feature Engineering · EDA · SQLite

DevSecOps: Ansible · CIS Benchmarks · CI/CD (Jenkins) · Black Duck · JFrog Artifactory · BlackBerry UEM

Observability: Grafana · Splunk · DataLore

Tools & Workflow: Git/GitHub · Android Studio · Jira · Confluence · Figma · PyCharm · VS Code · Agile/Scrum

WORK EXPERIENCE

Software Engineer · OneT Solutions · Singapore

Oct 2024 – Present

- Automated security audit and remediation workflows in Ansible across 8 enterprise device types (RHEL 9, Juniper OS, Oracle 19c, Cisco Firepower, Cisco ACI, Cisco DNA Center, Cisco SD-WAN, Cisco ISE), applying CIS Benchmark standards at scale.
- Performed API-driven testing by parsing JSON responses to verify audit accuracy and remediation outcomes, reinforcing backend engineering rigour in a security automation context.
- Engineered shell and PowerShell scripts aligned with vendor-provided security baselines to suit the use case of the client.
- Completed DevSecOps toolchain certifications across Black Duck, JFrog Artifactory, Grafana, and BlackBerry UEM.

Career development and independent learning in React and Java

May 2024 – Oct 2024

Software Consultant · FDM Group (Consulting) · Singapore

Jun 2023 – May 2024

Placement: OCBC Singapore — Mobile Software Engineer

Nov 2023 – Apr 2024

- Developed and integrated Bluetooth-enabled payment and multi-transfer features into the live OCBC mobile application (Kotlin / Android Studio), extending the PayNow flow and reducing transaction friction for millions of users.
- Presented the Bluetooth payment solution to C-suite stakeholders, including the CTO, at OCBC's Innovation Lab, translating technical decisions into business and UX outcomes for a non-technical executive audience.

- Pioneered a proof of concept for frictionless login via device-level credential caching, enabling direct dashboard access while preserving 2FA for sensitive operations, demonstrating security-conscious product thinking in a regulated environment.
- Authored unit tests for a recurring PayNow payments feature; resolved QA-tracked defects using Timber for logging and root cause analysis, contributing to on-time code merge and release.

FDM Internal — Full-Stack Java Engineer

Jun 2023 – Nov 2023

- Designed and built a full-stack Job Portal in Java, Spring Boot, Spring Security (role-based auth), Spring Data JPA, and MySQL, exposing and consuming RESTful APIs with full CRUD operations.
- Collaborated in an agile/scrum team to deliver a Forex Exchange Platform end-to-end, from Figma front-end design through Java/Spring Boot backend. Tracked across sprint cycles via GitHub, Confluence, and Jira.
- Built standalone Java projects to reinforce core engineering fundamentals, including a Unix-based file system management, currency conversion engine, elevator control system, and Hangman game.

Business Development Executive · Genplus Pte Ltd · Singapore

Dec 2019 – Jun 2023

- Managed client accounts in the renewable energy sector (PV and Energy Storage Systems), closing a deal valued at over S\$400,000 by owning the full sales cycle from lead qualification to delivery.
- Performed financial modelling and ROI analysis using HOMER simulation software, translating quantitative findings into stakeholder-facing investment recommendations.
- Conducted on-site feasibility assessments and coordinated cross-functional delivery teams, bridging technical and commercial stakeholders throughout project execution.

PROJECTS

Price Anomaly Detection System · Independent ML Project

May 2026 – Present

- Building an end-to-end ML system in PyTorch for price anomaly detection, covering neural network architecture design, time-series modelling, training loops, and model evaluation with NumPy and Pandas.
- Deploying via a FastAPI inference API and React Native front-end, applying production software engineering practices to a live AI system rather than staying at notebook-level experimentation.
- Designed to demonstrate the complete journey from raw data ingestion to a callable AI service, directly mirroring how AI features are integrated within product engineering teams at FinTechs.

Delivery Rating Prediction Pipeline · Independent ML Project

May 2026 – Jun 2026

- Applied Kruskal-Wallis, Chi-square, and Spearman/Pearson correlation tests to evaluate feature significance and inter-feature redundancy, in addition to domain knowledge and intuition on the raw dataset, to reduce 18 raw columns to 6 final features for a 5-class delivery rating classification problem.
- Trained and compared Logistic Regression, Random Forest, and Gradient Boosting models using Macro F1 as the primary metric and per-class recall for minority classes as the business interpretation metric, identifying Logistic Regression as the most suitable model for detecting poor delivery ratings despite lower overall accuracy.
- Orchestrated the full pipeline via a bash script, serialising trained models as joblib files for reproducible inference.

EDUCATION

Bachelor of Science (Economics)

Aug 2015 – Jun 2019

Singapore Management University (SMU)